Name: Jianyu FU

Position: Ph.D. Student

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The University of Tokyo

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Education

Ph.D. in Spatial Information Science
Graduate School of Frontier Sciences, The University of Tokyo, Japan
2025 – Present

- Research area: Urban perception, mobility data analysis, and AI-based cityscape modeling.
- Supervisor: Prof. Yoshihide Sekimoto
- Member of JST SPRING GX program (Grant No. JPMJSP2108)

M.Arch. in Architecture
Southeast University (Project 985), Nanjing, China
2021.09 – 2024.06

- Grade average: **90.1 / 100 (Rank 1 / 36)**
- Research direction: Urban Architecture, Urban Planning and Design, Urban Analytics, and Big Data Analysis.

B.Arch. in Architecture
Northeastern University (Project 985), Shenyang, China
2015.09 – 2020.06

- Undergraduate thesis focused on sustainable architectural design and human-environment interaction.

Publications & Conference Papers

- - **FU, J**., Wang, X., Ma, J., Zhao, C., Ogawa, Y., & Sekimoto, Y. (2026). *Evaluating urban street views through collective emotional perception from multimodal social media cityscape posts: An R-ECoT-based interpretable framework*.
  Landscape and Urban Planning (Elsevier). IF: 11.5 (First & Corresponding Author)
  DOI: 10.1016/j.landurbplan.2026.105822
- - **FU, J**., Xu, Y., Ma, J., & Sekimoto, Y. (2026). *Beyond Top-Down Urban Sensing: Cross-View Explanatory Asymmetry in Image-Estimated Streetscape Perception in Tokyo*.
  IEEE International Conference on Big Data (IEEE BigData 2026). Accepted for presentation
- - **FU, J**. (2025). *Linking urban morphology and human activity patterns: An empirical study based on mobility survey data from the Paris region*
  IEEE International Conference on Big Data (IEEE BigData 2025). Accepted for presentation
- - **FU, J**. (2025). *Urban Real Accessibility Evaluation Based on GPS Data: Integration of Space Syntax and Deep Learning*.
  Poster accepted for presentation at NetMob 2025, MIT & Inria, France.
- - **FU, J**. (2025). *Comprehensive Evaluation Method of Rooftop Views in High-Density Urban Areas: A Case Study of Guangzhou's Historic Districts*.
  Urban Design International (Springer Nature). (Second Author)
- - **FU, J**. (2024). *Research on Sustainable Renewal of Historic Blocks Based on Urban Catalyst Theory*.
  Applied and Computational Engineering.
- - **FU, J**. (2022, November). *Research on Urban Public Space Quality Evaluation Model Based on Scene Photo's Deep Learning*.
  2022 Computational Design Symposium and Annual Conference of the Computational Design Academic Committee of the Architectural Society of China.

- - **FU, J**. (2022, November). *Spatial Cognitive Preference of Traditional Village Nodes from the Perspective of Eye-Tracking Technology*.
  2022 Computational Design Symposium and Annual Conference of the Computational Design Academic Committee of the Architectural Society of China.

Research Interests

- Urban perception and emotion modeling
- Cityscape analytics and environmental cognition
- AI and deep learning in urban morphology
- Big data analysis for urban design and planning

Technical Skills

- **Programming:** Python (4+ years),
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, scikit-learn, Keras
- **Spatial Analysis & GIS:** QGIS, ArcGIS, Kepler.gl, PLATEAU 3D City Model
- **Data Processing:** Pandas, GeoPandas, OpenStreetMap & social media APIs
- **Visualization:** matplotlib, seaborn, plotly, Power BI

Professional Experience

Visiting Ph.D. Researcher
Department of Geography, The University of Hong Kong (Host: Prof. Bo Huang)
2026.08 – 2026.10

- Conducted collaborative research on urban mobility and human behavior, focusing on spatial patterns of individual travel-time burdens in Greater Paris.
- Analyzed the GNSS-based sanitized mobility dataset and urban spatial information to examine heterogeneity in observed travel experiences.
- Research manuscript: “When mobility continues at a cost: Individual travel-time burdens in Greater Paris” (under review).

Sponsored Researcher
SOMAR at ICPSR, University of Michigan
2025 – Present

- Appointed as a sponsored researcher/member of the University of Michigan with access to U-M computing services.
- Granted approved access to Meta Content Library API data through the SOMAR Virtual Data Enclave (VDE) under SOMARAPPLY-768 (SOMAR Data Application).
- Conduct research using publicly available Facebook and Instagram social media data in a secure computing environment for urban perception and collective emotion analysis.

Research Assistant
Sekimoto Laboratory, Center for Spatial Information Science (CSIS), The University of Tokyo
2025 – Present

Awards & Grants

- JST SPRING GX Fellowship, The University of Tokyo (2023–Present)
- NetMob 2025 Poster Track Selection (2025)
- Best Presentation Award at IEEE BigData 2025